Describe your professional, academic, or life experience and skills you have that enhance your ability to be an excellent contributing member to the SULI Program.

My technical depth, research experience in computational science, and interdisciplinary communication skills equip me to be a strong contributor to the SULI program. Throughout my academic journey, I have learned how to work independently in research environments, collaborate with people from diverse disciplines, and communicate complex scientific ideas clearly, which are paramount for being productive in a national laboratory setting.

Through research experience, coursework, and personal projects, I have become fluent in Python, C, C++, LaTeX, and I am comfortable using scientific computing packages such as NumPy, SciPy, Qiskit, Cirq, and Matplotlib. I have also worked in high-performance computing environments and gained familiarity with software development practices through open-source contributions, including version control, CI/CD workflows, and modular code design. These experiences have taught me how to write efficient, reproducible scientific code, which is essential for implementing computational methods across various scientific domains.

Beyond programming, I have learned how to approach and communicate complex scientific problems. I first became interested in computational methods through Monte Carlo algorithms during my first semester of college. I was intrigued by the versatility of these methods, and used them extensively in enhanced molecular dynamics sampling and optimizing quantum circuit simulator protocols. Learning LaTeX and practicing technical writing has strengthened my ability to communicate scientific ideas, which has been useful for preparing materials for conferences such as ACS Fall 2025 and for discussing results with experimental collaborators. Through these experiences, I have learned how to adapt my communication to audiences with different technical backgrounds, which is crucial for interdisciplinary research.

Working with computationally intensive systems such as large biomolecular systems or quantum circuits has taught me how to evaluate tradeoffs between numerical methods and choose appropriate approaches for different problems. Collaborating with postdocs and my principal investigators, and exploring relevant literature has helped me learn how to incorporate feedback and make decisions about the feasibility of different computational strategies.

My experience in quantum algorithms, molecular dynamics simulations, and computational physics has given me a broad perspective on both the utility and limitations of current computational methods. Conducting research in different fields has taught me about the importance of approaching problems from multiple angles and connecting methodologies across disciplines. I am eager to contribute my technical versatility and interdisciplinary knowledge to computational projects through the SULI program.